



Cognitive Load Transcript

Hello everyone, welcome to BITS Milani Digital. We will continue our learning journey for the course Data Visualisation and Storytelling. Today I will be talking about the topic Effective Visual Design and in this class we will be understanding the concepts of cognitive load, clutter and do a decluttering exercise.

The essence of this class is to make you appreciate the psychological principles that one has to adhere to or keep in mind while designing a visual and how to design visualisations that are less taxing on human perception and therefore are effectively perceived. We also see how cluttery can some images can get or how bad certain visualisations are and how to declutter or make them more effective. What is cognitive load? Simply put cognitive load is the amount of mental energy that you have to use to understand any information or visual in our case.

Lesser is the cognitive load, higher is the receptability and therefore retention. If you clutter your image present to heavy content or complex content the audience or the reader might have to spend a lot of time on the visual to understand what is the information being presented. Resultantly they might lose interest or at times worse misinterpret the data or information presented defeating the whole purpose of visualisation.

Therefore it is essential to optimally use the mental resources of our audience or effectively present our visuals so that the cognitive load is lower and the reception is high. Let us for example see the graphics that are on the slide. The first graph is very cluttered or has high cognitive load.

It is very interesting to see a 3d diagram but it defeats the purpose of conveying the information. In this case it might be sales or productions of different types of fruits, apples, oranges, lemons and bananas but it is not able to convey the actual production status because oranges are blocking the view on a 2d space. So whenever we are presenting this in a 2d space 3d diagrams will not make much of a sense because they are not able to appeal to the eye or at least require more mental effort to understand the information.

The same diagram if presented as a multiple bar could convey the information in a more effective and a less cognitively taxing format. Whenever you are designing the visual please place your audience in the centre and try to see the visual from their perspective and see whether it is easily perceptible. If that is the case then it conveys your information in an effective manner.

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